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description

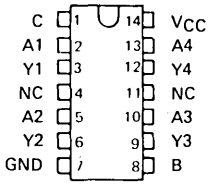
Operation of these monolithic 4-bit true/complement elements is controlled by the B and C inputs. With the B input low, the 4-bit binary input (A) is transferred to the output (Y) in either complementary form (with C low) or true form (with C high). When the B input is high, the output will be at the complementary level of the C input regardless of the levels of the data inputs.

These circuits are fully compatible for use with other TTL circuits. Input clamping diodes are provided to minimize transmission line effects and thereby simplify system design. Each input represents only one normalized series 54H/74H load, and full fan-out to 10 series 54H/74H loads is available from each of the outputs in the low-level condition.

Power dissipation is 270 mW typically with an average propagation delay of 14 ns from data inputs to output.

The SN54H87 is characterized for operation over the full military temperature range of −55°C to 125°C, and the SN74H87 is characterized for operation from 0°C to 70°C.

SN54H87 . . . J OR W PACKAGE
SN74H87 . . . J OR N PACKAGE
(TOP VIEW)



NC No internal connection

FUNCTION TABLE

CONTROL INPUTS		OUTPUTS			
B	C	Y1	Y2	Y3	Y4
L	L	$\overline{A1}$	$\overline{A2}$	$\overline{A3}$	$\overline{A4}$
L	H	A1	A2	A3	A4
H	L	H	H	H	H
H	H	L	L	L	L

H = high level, L = low level
A1, A2, A3, A4 = the level of the respective A input.

absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

Supply voltage, V_{CC} (see Note 1) . . . . . 7 V
Input voltage . . . . . 5.5 V
Operating free-air temperature range: SN54H87 Circuits . . . . . −55°C to 125°C
SN74H87 Circuits . . . . . 0°C to 70°C
Storage temperature range . . . . . −65°C to 150°C

NOTE 1: Voltage values are with respect to network ground terminal.

recommended operating conditions

	SN54H87			SN74H87			UNIT
	MIN	NOM	MAX	MIN	NOM	MAX	
Supply voltage, V_{CC}	4.5	5	5.5	4.75	5	5.25	V
High-level output current, I_{OH}			1			−1	mA
Low-level output current, I_{OL}			20			20	mA
Operating free-air temperature, T_A	−55		125	0		70	°C

PRODUCTION DATA
This document contains information current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

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